arXiv:2411.10083v3 [cs.CL] 4 Dec 2024

Xmodel-1.5: An 1B-scale Multilingual LLM

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Abstract

We introduce **Xmodel-1.5**, a 1-billion-parameter multilingual large language model pretrained on 2 trillion tokens, designed for balanced performance and scalability. Unlike most large models that use the BPE tokenizer, Xmodel-1.5 employs a custom unigram tokenizer with 65,280 tokens, optimizing both efficiency and accuracy. The model delivers competitive results across multiple languages, including Thai, Arabic, French, Chinese, and English, outperforming Alibaba's PolyLM-1.7B on respective evaluation datasets. Xmodel-1.5 excels in benchmarks like mMMLU and PIQA, and achieves state-of-the-art results in Thai. To support low-resource language research, we release **Xdata_Thai**, a Thai-specific evaluation dataset featuring unique linguistic challenges such as gendered particles and idioms. While the model demonstrates strong performance, there is still room for improvement in handling culturally specific nuances. We hope this work contributes to advancements in multilingual AI research. Models and code are publicly available at `https://github.com/XiaoduoAILab/XmodelLM-1.5`.

1 Introduction

The rapid globalization of communication has created an urgent need for advanced multilingual natural language processing (NLP) models that can bridge linguistic divides across regions. Traditional NLP models often struggle with less-represented languages, limiting their effectiveness in global applications. As cross-cultural interactions increase, there is a growing demand for AI systems that can understand and generate multiple languages with high accuracy and relevance.

In response to this challenge, Xiaoduo Technology's AI Lab has developed a 1-billion-parameter multilingual large model. Our model excels not only in widely spoken languages such as Chinese and English, but also in languages like Thai, Arabic, and French, demonstrating top-tier performance among models of similar scale. This work addresses the critical need for more inclusive AI systems capable of serving a wider array of linguistic and cultural contexts.

Beyond the model itself, we also contribute to the research community by open-sourcing a Thai evaluation dataset. This dataset, consisting of hundreds of questions annotated by students from Chulalongkorn University's School of Integrated Innovation, offers a valuable resource for future research in Thai language processing. These efforts highlight our commitment to advancing multilingual AI and improving the tools available for global communication and research. A detailed description of the dataset creation process is provided in Appendix 9.2.1.

2 Related Work on Multilingual Large Language Models

Multilingual large language models (LLMs) have gained significant attention in recent years, addressing the challenges of natural language processing across diverse languages. These models aim to generalize well across both high-resource and low-resource languages, offering a pathway for improved cross-lingual understanding and generation. Several notable models have contributed to this

area, including **XLM-R**, **mT5**, and **PolyLM**, which provide important benchmarks for multilingual AI development.

XLM-R (XLM-RoBERTa) [Conneau et al., 2020] is a widely-used pre-trained language model supporting over 100 languages. With parameter sizes ranging from 270M to 3.5B, XLM-R has set a high standard for natural language understanding tasks such as classification and question answering. Its robust generalization to low-resource languages has been a key factor in its success. The 1B parameter variant of XLM-R serves as an important benchmark for comparing models of similar scale, as it balances strong performance with computational efficiency.

mT5 (Multilingual T5) [Xue et al., 2021], developed by Google, is designed for both understanding and generation tasks across more than 100 languages. Ranging from 300M to 13B parameters, mT5 has shown impressive results, particularly in low-resource settings. The 1B parameter version is effective at handling complex cross-lingual tasks like machine translation and question answering, making it a suitable model for comparing performance on languages such as Thai, Arabic, and French.

PolyLM [Wei et al., 2023], developed by Alibaba DAMO Academy, is a more recent open-source multilingual model available in two sizes: 1.7B and 13B parameters. PolyLM incorporates bilingual data and utilizes a curriculum learning strategy to progressively introduce more non-English data during training. This approach improves its performance on lower-resource languages such as Thai and Indonesian. PolyLM has shown strong results across a wide range of multilingual tasks, often outperforming models like LLaMA and BLOOM on non-English languages.

3 Pretraining

This chapter details the pretraining process of Xmodel-1.5. We begin by introducing the sources and composition of our corpus, followed by an explanation of our preprocessing methods. Next, we describe the construction of our customized tokenizer. Finally, we outline the model architecture and training parameter configurations.

3.1 Training Data

Data Sourcing: Building upon the Chinese and English data from the first generation of Xmodel [Wang et al., 2024a], we focused on enhancing the model's performance in low-resource languages. To achieve this, we expanded the dataset by incorporating diverse multilingual data, specifically sourced from Multilang Wiki and CulturaX [Nguyen et al., 2023], as shown in Figure 1

The Wiki data was directly downloaded in 30 languages and preprocessed [CyberZHG, 2023] to remove unnecessary markers such as URLs. Similarly, CulturaX [Nguyen et al., 2023] provided data in 27 languages, with an emphasis on oversampling low-resource languages such as Mongolian (mn), Burmese (my), Nepali (ne), Khmer (km), Serbian (sr), and Tamil (ta), despite their smaller data volumes. To ensure data quality, we applied SimHash-based deduplication to the Wiki data, while leaving the CulturaX [Nguyen et al., 2023] data unchanged.

To further improve the model's proficiency in Chinese, we incorporated data from PTD [Wang et al., 2024b] and WanJuan [He et al., 2023], similar to the previous Xmodel. However, during the annealing phase, we transitioned the WanJuan dataset to a curated version that focuses on e-commerce domain data. Additionally, during the annealing phase, we introduced high-quality instructional data[Phatthiyaphaibun et al., 2023, AIRESEARCH, 2023, Phatthiyaphaibun, 2024, Lowphansirikul et al., 2020, Viriyayudhakorn and Polpanumas, 2021, Sawatphol, 2019] , collected with feedback from Thai colleagues. The proportions of this instructional data are provided in Table 1.

3.2 Tokenizer

For our multilingual large model, we used a unigram tokenizer [Kudo, 2018a] trained with SentencePiece [Kudo and Richardson, 2018], resulting in a vocabulary size of 65,280 tokens. This size strikes a balance between performance and efficiency for a 1 billion parameter model, enabling effective handling of diverse languages.

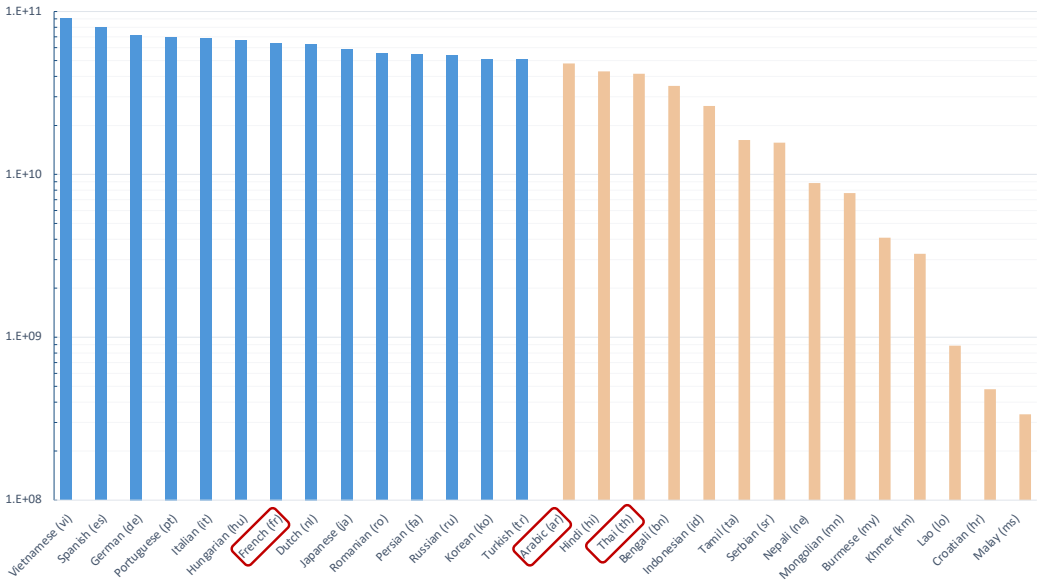

Figure 1: Total multilingual data tokens during the pretraining phase sourced from MultiWiki and CulturaX.

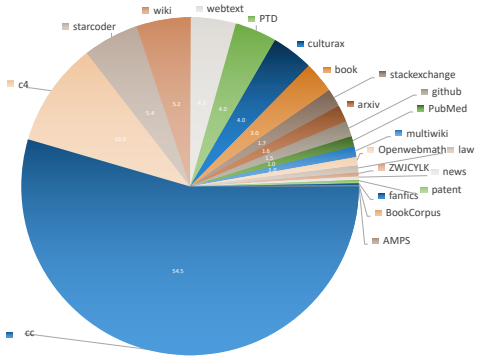

Figure 2: Data distribution during pretraining between 44,000 and 190,000 steps.

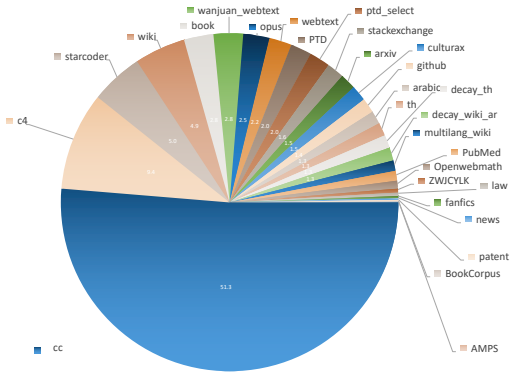

Figure 3: Data distribution during the decay phase.

We chose the unigram model [Kudo, 2018b] over the commonly used byte pair encoding (BPE) method [Bostrom and Durrett, 2020] because of its greater flexibility in handling rare and low-frequency tokens, as well as its faster training process. The unigram approach allows more adaptable word segmentation, capturing linguistic nuances and morphological variations, especially for low-resource languages with diverse word forms. While BPE is more memory-efficient, it requires much longer training times. For instance, in our experiments, a BPE model with a 128,000-token vocabulary took 60 hours and 300GB of memory for a 51.2GB dataset with 270 million lines, whereas the unigram model completed the same task in under 12 hours, though it required 1TB of memory due to its sampling process.

After several iterations of tokenizer design, we finalized a version with several key improvements:

Training Data Size and Distribution: The Xmodel-1.5 tokenizer was trained on a 50GB subset of the Xmodel-1.5 pre-training corpus, with additional industry-specific data to prepare the model for commercial applications. The data distribution was 50% English, 25% Chinese, 10% industry-specific, and 15% low-resource languages. No additional text normalization was applied.

Vocabulary Size: The vocabulary size was increased from 32,000 to 65,280 tokens to improve the model's ability to represent diverse languages and specialized terms. Numeric data was encoded by

Data Format	Dataset Name	Token Count	Category
instruct	klongklon	12,166,780	Literature
	thai_usembassy_th2en_prompt	2,000,250	News, Opus
	han-instruct-dataset-v4	2,711,072	Wiki QA, Legal, Opus, Web
	scb_mt_2020_th2en_prompt	277,208,596	Opus
	thai-wiktionary-prompt	3,989,318	Wiktionary
	prd_news_30112023	527,628,430	News
	th_iapp_wiki_qa_squad	6,373,522	Wiki QA
	thai-wiki-dataset-v3_processed_thaiwikibooks	8,040,228	Textbook
	oasst2_thai_top1_chat_format	93,256	Multi-task
	WangchanThaiInstruct_processed_dataset_thai	29,366,662	Medical, Finance, Retail, Legal
pretrain	vajirayana_filtered_tlc_content	103,112,882	News
	wiki_th	617,827,100	Wiki Docs
	tnhc	3,023,992	Arts
	thai_beginner_content	140,652	Opus
	combined_tlc_poems	18,955,824	Literature
	thai-oldbooks	38,930,572	Literature
	thai-it-books	579,196	Tech
	goethe-website	92,162	Culture
	thailand-policy-statements	1,152,378	Legal
	thai-constitution-corpus	1,551,736	Legal
	tlcv2.0_oa_processed_raw	12,687,966	Literature
	thai-financial-dataset	800,000,000	Financial

Table 1: Detailed Composition of Thai Decay Data.

splitting numbers into individual digits. Character coverage was set to 0.9999, with rare characters represented by UTF-8 bytes. To address out-of-vocabulary (OOV) issues, we enabled byte fallback mode and set a maximum token length of 16 to better handle Chinese phrases.

Whitespace Handling: Building on techniques from the LLaMA3 and InternLM2 tokenizers, we manually edited the vocabulary to include tokens for multiple consecutive spaces, improving compression rates for code data. Inspired by MAP-NEO [Zhang et al., 2024a], we disabled SentencePiece's default option to remove extra spaces by setting `-remove_extra_whitespaces=false`, which solved a formatting issue observed in Xmodel-LM's code generation [Wang et al., 2024a]. This adjustment has been incorporated into Xmodel-1.5.

These improvements make the Xmodel-1.5 tokenizer particularly well-suited for multilingual tasks, including those involving low-resource languages and code generation, by balancing language coverage, representational capacity, and data processing efficiency. A comparison of the Xmodel-1.5 tokenizer with other widely used tokenizers is shown in Table 2, where our tokenizer demonstrates impressive compression rates despite its relatively small size.

Tokenizer	Vocab Size	Compression Rate ↓
LLaMA 3	128,000	0.3823
LLaMA 2	32,000	0.7524
InternLM 2	103,168	0.4124
Baichuan 2	125,696	0.4103
Xmodel-1.5	65,280	**0.3800**

Table 2: Comparison of vocabulary size and text compression rate of Xmodel-1.5's tokenizer with other models. Lower values indicate better compression.

3.3 Model architecture

We built upon the architecture of the first generation of the Xmodel [Wang et al., 2024a] with the following details:

Hidden size	Intermediate size	Attention heads	KV heads	Layers	Context Len
2048	5632	32	4	24	4096

Table 3: Detailed settings of Xmodel-1.5.

Rotary Positional Embedding. We integrate rotary positional embeddings (RoPE) [Su et al., 2023] at each layer of the network. Additionally, to improve the model's long-context understanding capability, we adopted the approach from [Xiong et al., 2023] and adjusted the RoPE base from 10,000 to 500,000.

RMSNorm. To enhance training stability, we utilize the RMSNorm [Zhang and Sennrich, 2019] function to normalize the input of each transformer sub-layer, without normalizing the output. Linear layers do not incorporate bias, and word embeddings are not tied.

SwiGLU. We replace the conventional ReLU non-linearity with the SwiGLU [Shazeer, 2020] activation function to optimize performance.

Grouped-query attention. For efficient training and inference, we employ grouped-query attention (GQA) [Ainslie et al., 2023], which incorporates 32 attention heads and 4 KV heads.

3.4 Training

Training is performed on a single node using 7 out of the 8 available H800 GPUs. To enhance efficiency and maximize Model FLOPS Utilization (MFU), we utilize Distributed Data Parallel (DDP) and FlashAttention-V2.

We apply cumulative gradient updating, setting a mini-batch size of 4 and using 30 gradient accumulation steps per GPU, resulting in a global batch size of 840 with a sequence length of 4096. This configuration produces a total of 3,440,640 tokens per iteration. Training is conducted over 600,000 iterations, yielding a total token count of 2,064,384,000,000.

We optimize using the AdamW optimizer with a peak learning rate of 6e-4. The learning rate linearly ramps up from zero to the peak over the first 2000 updates, then gradually decreases to 2e-5 following a cosine schedule. At 478K iterations, we introduce an exponential decay to further reduce the learning rate. Throughout training, the effective batch size is approximately 3.5 million tokens, with a weight decay rate of 0.1. We also apply gradient clipping with a threshold of 1.0 to control gradient values.

Data allocation evolves throughout training, with the proportion of multilingual data increasing from 5% to 10%, with the initial data ratio as illustrated in Figure 2

Refer to Figure 4 for training logs, which include trend graphs depicting the progression of training and validation losses as the total token count increases. We use OpenWebText2 [Gokaslan et al., 2019], a dataset excluded from training, as the validation set to measure validation loss.

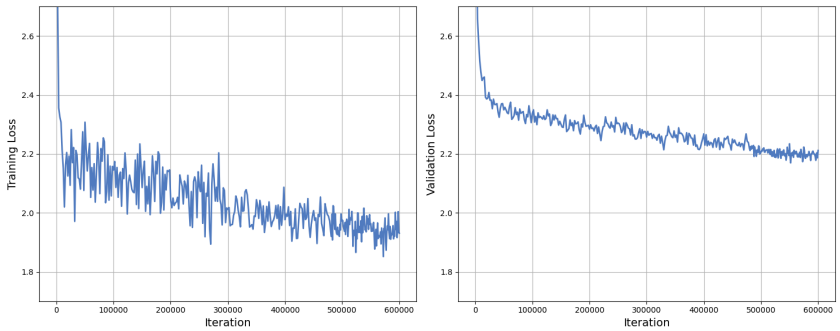

Figure 4: The trend of training and validation loss during pretraining.

4 Post-training

Instruction fine-tuning has become a pivotal approach for improving large pre-trained language models on instruction-based tasks. To enable our model to perform well on e-commerce Retrieval-Augmented Generation (RAG) tasks, we trained an instruction-tuned model using our foundational architecture. Instead of restricting loss computation and backpropagation to just the answer outputs, we utilized the full instruction dataset, resulting in more effective training [Shi et al., 2024].

Our fine-tuning setup included a learning rate of 6e-5, weight decay of 0.1, a warmup ratio of 0.03, a context length of 4096 tokens, and a batch size of 120, with a cosine annealing scheduler. We constructed the dataset progressively, starting from simple tasks and advancing to more complex ones, integrating the RAG and RAFT datasets [Zhang et al., 2024c] toward the end. The dataset comprised Belle (56.04%) [Ji et al., 2023], infinity-instruct-subject (34.25%), infinity-commonsense (5.97%), RAG_mixed (2.97%), and RAFT_mixed (0.77%).

The model's RAG performance in the e-commerce domain achieved a satisfaction rate of 92.47%, as evaluated by GPT-4o mini on our e-commerce evaluation dataset.

5 Evaluation

5.1 Evaluating the Pretrained Model

Baseline Models To ensure a fair comparison, we selected several popular decoder-only architecture models, each with approximately 1 billion parameters. Specifically, we compare Xmodel-1.5 with the following models: OPT [Zhang et al., 2022], Pythia [Biderman et al., 2023], TinyLLaMA [Zhang et al., 2024b], MobileLLaMA [Thawakar et al., 2024], H2O-danube [Singer et al., 2024], InternLM2 [Cai et al., 2024] and Qwen2.5 [Hui et al., 2024].

Commonsense reasoning tasks We evaluate our models using the Language Model Evaluation Harness [Gao et al., 2023], which includes tasks such as ARC-Challenge [Clark et al., 2018b], ARC-Easy [Clark et al., 2018b], Boolq [Clark et al., 2019], HellaSwag [Zellers et al., 2019], OpenBookQA [Mihaylov et al., 2018], PiQA [Bisk et al., 2019], SciQ [Welbl et al., 2017], and Winogrande [Sakaguchi et al., 2021]. To ensure fairness and reproducibility, all evaluation metrics were computed in the same environment. Notably, we report raw accuracy metrics, rather than normalized ones. The evaluation results, presented in Table 4, show that our model, Xmodel-1.5, outperforms several baseline models, especially surpassing TinyLlama across multiple evaluation metrics.

Model	**ARC-c**	**ARC-e**	**Boolq**	**HS.**	**OB.**	**PiQA**	**SciQ**	**Wino.**	**Avg**
OPT-1.3B	23.29	57.03	57.80	41.52	23.20	71.71	84.30	59.59	52.32
Pythia-1.4B	25.60	57.58	60.34	39.81	20.20	71.06	85.20	56.20	53.38
TinyLLaMA-3T-1.1B	27.82	60.31	57.83	44.98	21.80	73.34	88.90	59.12	54.26
MobileLLaMA-1.4B	26.28	61.32	57.92	42.87	23.60	71.33	87.40	58.25	53.60
InternLM2-1.8B	37.54	70.20	69.48	46.52	24.40	75.57	93.90	65.67	60.41
Qwen2.5-1.5B	40.36	74.83	73.27	50.09	31.40	75.95	94.90	63.06	62.98
Xmodel-1.5-1B	28.92	64.31	62.78	45.94	22.20	72.20	89.10	60.62	55.76

Table 4: Performance on commonsense reasoning tasks. Models marked in green perform worse than Xmodel-1.5, while models marked in red perform better than Xmodel-1.5.

Multilingual ability In addition to evaluating the model's proficiency in English, we also assessed its multilingual capabilities. Specifically, our evaluation included the following tasks:

- ARC [Clark et al., 2018a]: This dataset consists of 7,787 science exam questions from various sources, including questions provided by a research partner affiliated with AI2. We used the Chinese-translated version of this dataset for our evaluation.
- XCOPA [Ponti et al., 2020]: Designed to test how well machine learning models transfer commonsense reasoning across different languages, XCOPA is a translated and reannotated version of the English COPA [Gordon et al., 2011] and includes 11 languages from different language families and regions worldwide.

- PIQA_AR [Almazrouei et al., 2023]: This is the Arabic version of the PIQA dataset, designed to evaluate physical commonsense reasoning in models, translated by AlGhafa.

- Belebele_tha_thai [Bandarkar et al., 2023]: A Thai subset of the Belebele Benchmark, which assesses multilingual models' reading comprehension through multiple-choice questions based on FLORES-200 passages.

- mMMLU[Hendrycks et al., 2021]: This benchmark evaluates models across 57 tasks, including math, history, and law, highlighting knowledge gaps, particularly in areas like morality and law.

- mHellaswag[Hendrycks et al., 2021]: A machine-translated version of the HellaSwag [Zellers et al., 2019] dataset, which includes multiple-choice questions to test commonsense reasoning.

The evaluation results are shown in Figure 5, with further details on the model's performance evolution provided in the case study section 6.2.

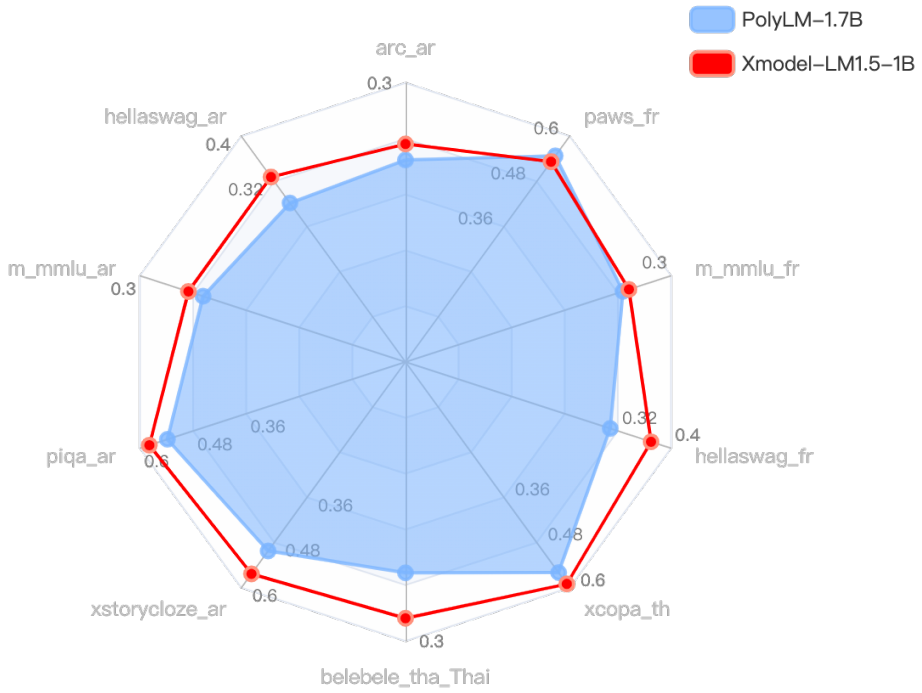

Figure 5: Comparison of performance in multilingual tasks between PolyLM 1.7B and Xmodel-1.5 1B

5.2 Evaluation of the Instruction Model

To evaluate the performance of our instruction model, we conducted a series of standard assessments, including ifeval [Zhou et al., 2023], which tests various aspects of language understanding and instruction-following abilities, as well as MT-Bench [Bai et al., 2024], a fine-grained benchmark designed to evaluate large language models in multi-turn dialogues. The results, summarized in Table 5, highlight the model's strengths and weaknesses in following instructions and managing dialogue complexity.

Model	**IFEval**	**MT-Bench**
TinyLlama-1.1B-Chat-v1.0	5.96	3.46
Qwen2.5-1.5B-Instruct	42.5	N/A
StableLM-2-zephyr-1.6B	32.79	5.42
H2O-Danube-1.8B-Chat	15.16	5.52
InternLM2-Chat-1.8B	23.87	4.94
Gemma-2B	20.38	5.19
Phi-2-2.7B	27.39	4.29
PolyLM-Chat-13B	16.27	N/A
Xmodel-1.5-Instruct-1B	3.7	5.06

Table 5: Performance on instruction following and multi-turns chat tasks.

6 Case Study

6.1 Evaluation Insights from Chulalongkorn Collaboration

We collaborated with Chulalongkorn University to develop a user-based evaluation and annotation interface, as shown in Figure 9. This interface enabled students to perform ad hoc testing and rate model outputs, with feedback and suggested corrections stored in a backend database.

Through this collaboration, several key observations were made:

- The model performed effectively in structured tasks such as e-commerce Q&A, where responses were described as concise and clear by native speakers (Figure 10).
- However, challenges were identified in handling Thai-specific linguistic nuances:
  - **Gendered Language**: The model struggled to differentiate gendered particles, an issue also observed in other state-of-the-art models (Figure 11).
  - **Time and Numerical Expressions**: Unique ways of expressing time and numerical data in Thai often resulted in translation errors and misrepresentations (Figure 12).

These findings highlight areas for further refinement, particularly in gender differentiation, politeness, and handling culturally specific expressions. A detailed analysis, including examples of good and poor cases, is provided in Appendix 9.1.

6.2 Evolution of the Model's Performance

We monitored and recorded the model's performance on the multilingual benchmark throughout the pretraining process, as shown in Figure 6, 7, 8. As training progressed, it became evident that the performance of Xmodel-1.5 consistently improved, ultimately surpassing PolyLM-1.7B [Wei et al., 2023] across multiple tasks.

7 Conclusions

In summary, our 1-billion-parameter multilingual large language model represents a valuable step forward in the exploration of cross-linguistic small language models (SLMs). It demonstrates strong performance across languages such as Thai, Arabic, and French, showcasing its potential to significantly enhance multilingual AI systems and cross-cultural communication. These results validate the effectiveness of our approach and highlight the model's promise in addressing low-resource language challenges and domain-specific applications, including e-commerce. While encouraged by these achievements, we recognize opportunities for further refinement, particularly in handling complex linguistic nuances such as gendered expressions, temporal data, and culturally specific idioms. This work provides a foundation for ongoing advancements in multilingual AI, paving the way for more inclusive, accurate, and culturally aligned systems. We are confident that these contributions will play a pivotal role in bridging language barriers and fostering greater understanding across diverse linguistic and cultural landscapes.

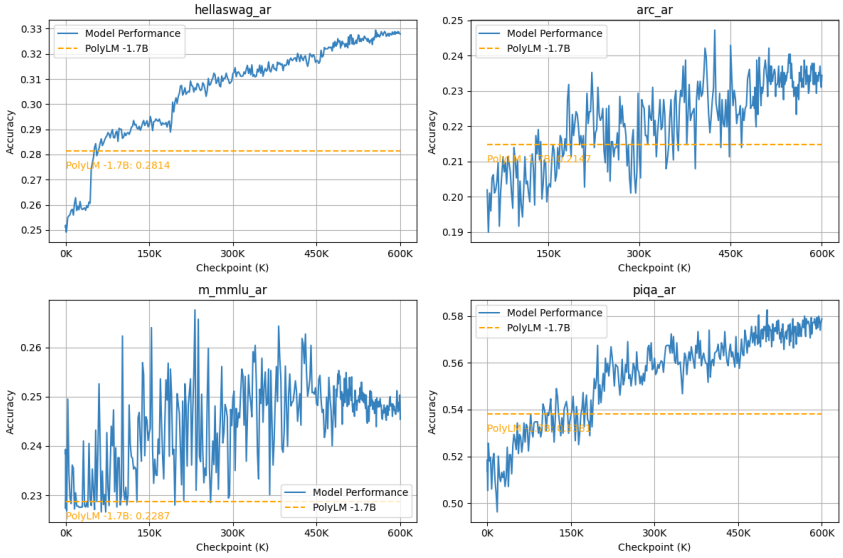

Figure 6: Evolution of performance in Arabic bench during pre-training

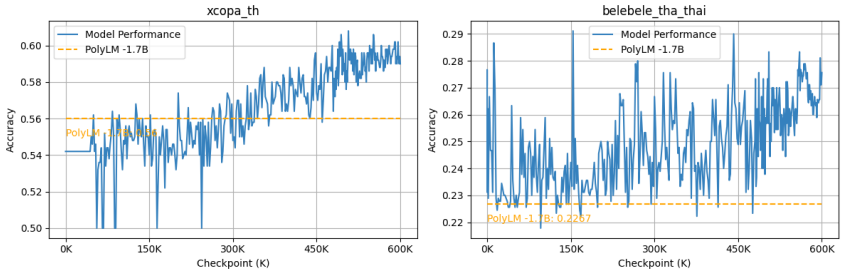

Figure 7: Evolution of performance in Thai bench during pre-training

8 Acknowledgments

We would like to express our sincere gratitude to Dr. Pietro Borsano and students from the School of Integrated Innovation at Chulalongkorn University in Thailand for their invaluable contributions to the Thai large language model collaboration project. Their enthusiasm and dedication, particularly in annotating the Thai evaluation dataset, were instrumental to the success of this work. We also extend our thanks to our colleagues at Xiaoduo Technology, including Xu Zongliang, Shi Yu, Chen Nanxi, and Chen Xinjie, for their efforts in organizing and supporting the project. Their commitment and teamwork were crucial to making this achievement possible. Finally, we would like to thank Qin Shuo for his helpful suggestions on improving the clarity of our writing.

References

Joshua Ainslie, James Lee-Thorp, Michiel de Jong, Yury Zemlyanskiy, Federico Lebrón, and Sumit Sanghai. Gqa: Training generalized multi-query transformer models from multi-head checkpoints, 2023.

AIRESEARCH. Wangchanthaiinstruct: A thai instruction-following dataset. https://huggingface.co/datasets/airesearch/WangchanThaiInstruct, 2023. URL https://huggingface.co/datasets/airesearch/WangchanThaiInstruct. Accessed: 2024-11-15.

Ebtesam Almazrouei, Ruxandra Cojocaru, Michele Baldo, Quentin Malartic, Hamza Alobeidli, Daniele Mazzotta, Guilherme Penedo, Giulia Campesan, Mugariya Farooq, Maitha Alhammadi, Julien Launay, and Badreddine Noune. AlGhafa evaluation benchmark for Arabic language models. In Hassan Sawaf, Samhaa El-Beltagy, Wajdi Zaghouani, Walid Magdy, Ahmed Abdelali, Nadi Tomeh, Ibrahim Abu Farha, Nizar Habash, Salam Khalifa, Amr Keleg, Hatem Haddad, Imed Zitouni, Khalil Mrini, and Rawan Almatham, editors, *Proceedings of ArabicNLP 2023*, pages 244–275, Singapore (Hybrid), December 2023. Association for Computational Linguistics. doi: 10.18653/v1/2023.arabicnlp-1.21. URL https://aclanthology.org/2023.arabicnlp-1.21.

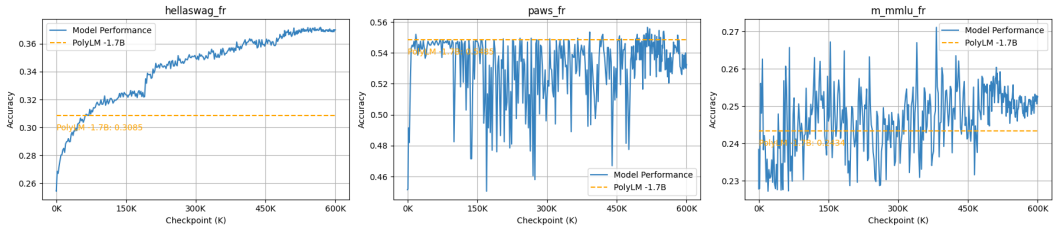

Figure 8: Evolution of performance in French bench during pre-training

Ge Bai, Jie Liu, Xingyuan Bu, Yancheng He, Jiaheng Liu, Zhanhui Zhou, Zhuoran Lin, Wenbo Su, Tiezheng Ge, Bo Zheng, and Wanli Ouyang. Mt-bench-101: A fine-grained benchmark for evaluating large language models in multi-turn dialogues. In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 7421–7454. Association for Computational Linguistics, 2024. doi: 10.18653/v1/2024.acl-long.401. URL http://dx.doi.org/10.18653/v1/2024.acl-long.401.

Lucas Bandarkar, Davis Liang, Benjamin Muller, Mikel Artetxe, Satya Narayan Shukla, Donald Husa, Naman Goyal, Abhinandan Krishnan, Luke Zettlemoyer, and Madian Khabsa. The belebele benchmark: a parallel reading comprehension dataset in 122 language variants, 2023.

Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O'Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling, 2023.

Yonatan Bisk, Rowan Zellers, Ronan Le Bras, Jianfeng Gao, and Yejin Choi. Piqa: Reasoning about physical commonsense in natural language. In *AAAI Conference on Artificial Intelligence*, 2019. URL https://api.semanticscholar.org/CorpusID:208290939.

Kaj Bostrom and Greg Durrett. Byte pair encoding is suboptimal for language model pretraining, 2020. URL https://arxiv.org/abs/2004.03720.

Zheng Cai, Maosong Cao, Haojiong Chen, Kai Chen, Keyu Chen, Xin Chen, Xun Chen, Zehui Chen, Zhi Chen, Pei Chu, Xiaoyi Dong, Haodong Duan, Qi Fan, Zhaoye Fei, Yang Gao, Jiaye Ge, Chenya Gu, Yuzhe Gu, Tao Gui, Aijia Guo, Qipeng Guo, Conghui He, Yingfan Hu, Ting Huang, Tao Jiang, Penglong Jiao, Zhenjiang Jin, Zhikai Lei, Jiaxing Li, Jingwen Li, Linyang Li, Shuaibin Li, Wei Li, Yining Li, Hongwei Liu, Jiangning Liu, Jiawei Hong, Kaiwen Liu, Kuikun Liu, Xiaoran Liu, Chengqi Lv, Haijun Lv, Kai Lv, Li Ma, Runyuan Ma, Zerun Ma, Wenchang Ning, Linke Ouyang, Jiantao Qiu, Yuan Qu, Fukai Shang, Yunfan Shao, Demin Song, Zifan Song, Zhihao Sui, Peng Sun, Yu Sun, Huanze Tang, Bin Wang, Guoteng Wang, Jiaqi Wang, Jiayu Wang, Rui Wang, Yudong Wang, Ziyi Wang, Xingjian Wei, Qizhen Weng, Fan Wu, Yingtong Xiong, Chao Xu, Ruiliang Xu, Hang Yan, Yirong Yan, Xiaogui Yang, Haochen Ye, Huaiyuan Ying, Jia Yu, Jing Yu, Yuhang Zang, Chuyu Zhang, Li Zhang, Pan Zhang, Peng Zhang, Ruijie Zhang, Shuo Zhang, Songyang Zhang, Wenjian Zhang, Wenwei Zhang, Xingcheng Zhang, Xinyue Zhang, Hui Zhao, Qian Zhao, Xiaomeng Zhao, Fengzhe Zhou, Zaida Zhou, Jingming Zhuo, Yicheng Zou, Xipeng Qiu, Yu Qiao, and Dahua Lin. Internlm2 technical report, 2024.

Christopher Clark, Kenton Lee, Ming-Wei Chang, Tom Kwiatkowski, Michael Collins, and Kristina Toutanova. Boolq: Exploring the surprising difficulty of natural yes/no questions, 2019.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge. *ArXiv*, abs/1803.05457, 2018a.

Peter Clark, Isaac Cowhey, Oren Etzioni, Tushar Khot, Ashish Sabharwal, Carissa Schoenick, and Oyvind Tafjord. Think you have solved question answering? try arc, the ai2 reasoning challenge, 2018b.

Alexis Conneau, Kartikay Khandelwal, Naman Goyal, Vishrav Chaudhary, Guillaume Wenzek, Francisco Guzmán, Edouard Grave, Myle Ott, Luke Zettlemoyer, and Veselin Stoyanov. Unsupervised cross-lingual representation learning at scale, 2020. URL https://arxiv.org/abs/1911.02116.

CyberZHG. wiki-dump-reader. https://github.com/CyberZHG/wiki-dump-reader, 2023. Accessed: 2024-10-23.

Leo Gao, Jonathan Tow, Baber Abbasi, Stella Biderman, Sid Black, Anthony DiPofi, Charles Foster, Laurence Golding, Jeffrey Hsu, Alain Le Noac'h, Haonan Li, Kyle McDonell, Niklas Muennighoff, Chris Ociepa, Jason Phang, Laria Reynolds, Hailey Schoelkopf, Aviya Skowron, Lintang Sutawika, Eric Tang, Anish Thite, Ben Wang, Kevin Wang, and Andy Zou. A framework for few-shot language model evaluation, 12 2023. URL https://zenodo.org/records/10256836.

Aaron Gokaslan, Vanya Cohen, Ellie Pavlick, and Stefanie Tellex. Openwebtext corpus. http://Skylion007.github.io/OpenWebTextCorpus, 2019.

Andrew S. Gordon, Zornitsa Kozareva, and Melissa Roemmele. Choice of plausible alternatives: An evaluation of commonsense causal reasoning. In *AAAI Spring Symposium: Logical Formalizations of Commonsense Reasoning*, 2011. URL https://api.semanticscholar.org/CorpusID:434646.

Conghui He, Zhenjiang Jin, Chao Xu, Jiantao Qiu, Bin Wang, Wei Li, Hang Yan, Jiaqi Wang, and Dahua

Lin. Wanjuan: A comprehensive multimodal dataset for advancing english and chinese large models, 2023.

Dan Hendrycks, Collin Burns, Steven Basart, Andy Zou, Mantas Mazeika, Dawn Song, and Jacob Steinhardt. Measuring massive multitask language understanding, 2021. URL https://arxiv.org/abs/2009.03300.

Binyuan Hui, Jian Yang, Zeyu Cui, Jiaxi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Kai Dang, An Yang, Rui Men, Fei Huang, Xingzhang Ren, Xuancheng Ren, Jingren Zhou, and Junyang Lin. Qwen2.5-coder technical report, 2024. URL https://arxiv.org/abs/2409.12186.

Yunjie Ji, Yong Deng, Yan Gong, Yiping Peng, Qiang Niu, Lei Zhang, Baochang Ma, and Xiangang Li. Exploring the impact of instruction data scaling on large language models: An empirical study on real-world use cases. *arXiv preprint arXiv:2303.14742*, 2023.

Matt Gardner Johannes Welbl, Nelson F. Liu. Crowdsourcing multiple choice science questions, 2017.

Taku Kudo. Subword regularization: Improving neural network translation models with multiple subword candidates, 2018a.

Taku Kudo. Subword regularization: Improving neural network translation models with multiple subword candidates, 2018b. URL https://arxiv.org/abs/1804.10959.

Taku Kudo and John Richardson. Sentencepiece: A simple and language independent subword tokenizer and detokenizer for neural text processing, 2018.

Lalita Lowphansirikul, Charin Polpanumas, Attapol T Rutherford, and Sarana Nutanong. scb-mt-en-th-2020: A large english-thai parallel corpus. *arXiv preprint arXiv:2007.03541*, 2020.

Todor Mihaylov, Peter Clark, Tushar Khot, and Ashish Sabharwal. Can a suit of armor conduct electricity? a new dataset for open book question answering, 2018.

Thuat Nguyen, Chien Van Nguyen, Viet Dac Lai, Hieu Man, Nghia Trung Ngo, Franck Dernoncourt, Ryan A. Rossi, and Thien Huu Nguyen. Culturax: A cleaned, enormous, and multilingual dataset for large language models in 167 languages, 2023.

Wannaphong Phatthiyaphaibun. Thai tnhc2 books, 2024. URL https://doi.org/10.5281/zenodo.10783421.

Wannaphong Phatthiyaphaibun, Korakot Chaovavanich, Charin Polpanumas, Arthit Suriyawongkul, Lalita Lowphansirikul, Pattarawat Chormai, Peerat Limkonchotiwat, Thanathip Suntorntip, and Can Udomcharoenchaikit. PyThaiNLP: Thai natural language processing in Python. In Liling Tan, Dmitrijs Milajevs, Geeticka Chauhan, Jeremy Gwinnup, and Elijah Rippeth, editors, *Proceedings of the 3rd Workshop for Natural Language Processing Open Source Software (NLP-OSS 2023)*, pages 25–36, Singapore, Singapore, December 2023. Empirical Methods in Natural Language Processing. URL https://aclanthology.org/2023.nlposs-1.4.

Edoardo M. Ponti, Goran Glavaš, Olga Majewska, Qianchu Liu, Ivan Vulić, and Anna Korhonen. XCOPA: A multilingual dataset for causal commonsense reasoning. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2020. URL https://ducdauge.github.io/files/xcopa.pdf.

Keisuke Sakaguchi, Ronan Le Bras, Chandra Bhagavatula, and Yejin Choi. Winogrande: an adversarial winograd schema challenge at scale. *Commun. ACM*, 64(9):99–106, aug 2021. ISSN 0001-0782. doi: 10.1145/3474381. URL https://doi.org/10.1145/3474381.

Jitkapat Sawatphol. Thai literature corpora. urlhttps://attapol.github.io/tlc.html, 2019.

Noam Shazeer. Glu variants improve transformer, 2020.

Zhengyan Shi, Adam X. Yang, Bin Wu, Laurence Aitchison, Emine Yilmaz, and Aldo Lipani. Instruction tuning with loss over instructions, 2024. URL https://arxiv.org/abs/2405.14394.

Philipp Singer, Pascal Pfeiffer, Yauhen Babakhin, Maximilian Jeblick, Nischay Dhankhar, Gabor Fodor, and Sri Satish Ambati. H2o-danube-1.8b technical report, 2024.

Jianlin Su, Yu Lu, Shengfeng Pan, Ahmed Murtadha, Bo Wen, and Yunfeng Liu. Roformer: Enhanced transformer with rotary position embedding, 2023.

Omkar Thawakar, Ashmal Vayani, Salman Khan, Hisham Cholakal, Rao M. Anwer, Michael Felsberg, Tim Baldwin, Eric P. Xing, and Fahad Shahbaz Khan. Mobillama: Towards accurate and lightweight fully transparent gpt, 2024.

Kobkrit Viriyayudhakorn and Charin Polpanumas. iapp_wiki_qa_squad, February 2021. URL https://doi.org/10.5281/zenodo.4539916.

Yichuan Wang, Yang Liu, Yu Yan, Qun Wang, Xucheng Huang, and Ling Jiang. Xmodel-lm technical report, 2024a. URL https://arxiv.org/abs/2406.02856.

Zihan Wang, Xinzhang Liu, Shixuan Liu, Yitong Yao, Yuyao Huang, Zhongjiang He, Xuelong Li, Yongxiang Li, Zhonghao Che, Zhaoxi Zhang, Yan Wang, Xin Wang, Luwen Pu, Huihan Xu, Ruiyu Fang, Yu Zhao, Jie Zhang, Xiaomeng Huang, Zhilong Lu, Jiaxin Peng, Wenjun Zheng, Shiquan Wang, Bingkai Yang, Xuewei he, Zhuoru Jiang, Qiyi Xie, Yanhan Zhang, Zhongqiu Li, Lingling Shi, Weiwei Fu, Yin Zhang, Zilu Huang, Sishi Xiong, Yuxiang Zhang, Chao Wang, and Shuangyong Song. Telechat technical report, 2024b.

Xiangpeng Wei, Haoran Wei, Huan Lin, Tianhao Li, Pei Zhang, Xingzhang Ren, Mei Li, Yu Wan, Zhiwei Cao, Binbin Xie, Tianxiang Hu, Shangjie Li, Binyuan Hui, Bowen Yu, Dayiheng Liu, Baosong Yang, Fei Huang, and Jun Xie. Polylm: An open source polyglot large language model, 2023. URL https://arxiv.org/abs/2307.06018.

Johannes Welbl, Nelson F. Liu, and Matt Gardner. Crowdsourcing multiple choice science questions, 2017.

Wenhan Xiong, Jingyu Liu, Igor Molybog, Hejia Zhang, Prajjwal Bhargava, Rui Hou, Louis Martin, Rashi Rungta, Karthik Abinav Sankararaman, Barlas Oguz, Madian Khabsa, Han Fang, Yashar Mehdad, Sharan Narang, Kshitiz Malik, Angela Fan, Shruti Bhosale, Sergey Edunov, Mike Lewis, Sinong Wang, and Hao Ma. Effective long-context scaling of foundation models, 2023. URL https://arxiv.org/abs/2309.16039.

Linting Xue, Noah Constant, Adam Roberts, Mihir Kale, Rami Al-Rfou, Aditya Siddhant, Aditya Barua, and Colin Raffel. mt5: A massively multilingual pre-trained text-to-text transformer, 2021. URL https://arxiv.org/abs/2010.11934.

Rowan Zellers, Ari Holtzman, Yonatan Bisk, Ali Farhadi, and Yejin Choi. Hellaswag: Can a machine really finish your sentence?, 2019.

Biao Zhang and Rico Sennrich. *Root mean square layer normalization*. Curran Associates Inc., Red Hook, NY, USA, 2019.

Ge Zhang, Scott Qu, Jiaheng Liu, Chenchen Zhang, Chenghua Lin, Chou Leuang Yu, Danny Pan, Esther Cheng, Jie Liu, Qunshu Lin, Raven Yuan, Tuney Zheng, Wei Pang, Xinrun Du, Yiming Liang, Yinghao Ma, Yizhi Li, Ziyang Ma, Bill Lin, Emmanouil Benetos, Huan Yang, Junting Zhou, Kaijing Ma, Minghao Liu, Morry Niu, Noah Wang, Quehry Que, Ruibo Liu, Sine Liu, Shawn Guo, Soren Gao, Wangchunshu Zhou, Xinyue Zhang, Yizhi Zhou, Yubo Wang, Yuelin Bai, Yuhan Zhang, Yuxiang Zhang, Zenith Wang, Zhenzhu Yang, Zijian Zhao, Jiajun Zhang, Wanli Ouyang, Wenhao Huang, and Wenhu Chen. Map-neo: Highly capable and transparent bilingual large language model series, 2024a. URL https://arxiv.org/abs/2405.19327.

Peiyuan Zhang, Guangtao Zeng, Tianduo Wang, and Wei Lu. Tinyllama: An open-source small language model, 2024b.

Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, et al. Opt: Open pre-trained transformer language models. *arXiv preprint arXiv:2205.01068*, 2022.

Tianjun Zhang, Shishir G. Patil, Naman Jain, Sheng Shen, Matei Zaharia, Ion Stoica, and Joseph E. Gonzalez. Raft: Adapting language model to domain specific rag, 2024c. URL https://arxiv.org/abs/2403.10131.

Jeffrey Zhou, Tianjian Lu, Swaroop Mishra, Siddhartha Brahma, Sujoy Basu, Yi Luan, Denny Zhou, and Le Hou. Instruction-following evaluation for large language models, 2023. URL https://arxiv.org/abs/2311.07911.

9 Appendix

9.1 Project Collaboration with Chulalongkorn University

This section presents the results and observations from our project collaboration with Chulalongkorn University.

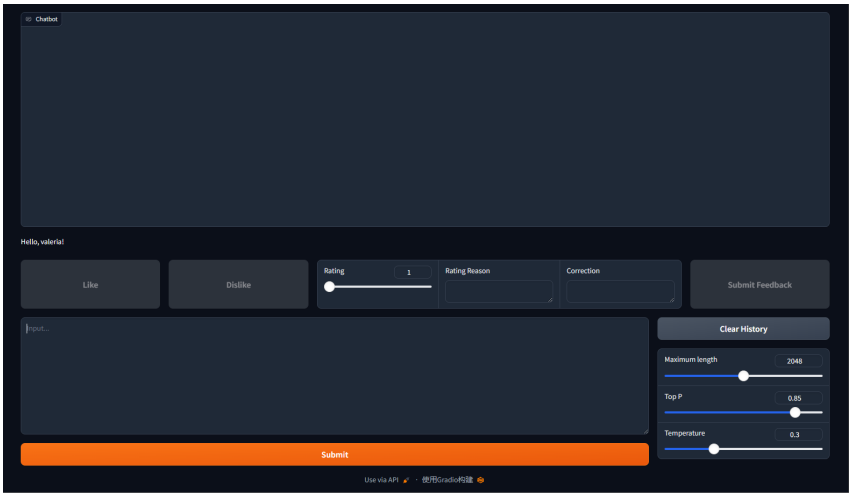

Figure 9: Annotation interface for students at Chulalongkorn University

We found that Xmodel-1.5 performs effectively in the e-commerce QA context. Feedback from students at Chulalongkorn University described it as "concise and easy to understand." The Thai text in the Figure 10 translates as follows:

User: To which account will the refund be issued?
Assistant: The refund will be issued to the account used for the purchase.
User: What is the procedure if I want to exchange the product?
Assistant: To exchange a product, first contact the seller or their representative to inform them of your intention. You will then receive a reference number. After that, reach out again to the seller or representative, provide details of the product you wish to exchange, and await the new product based on the reference number.

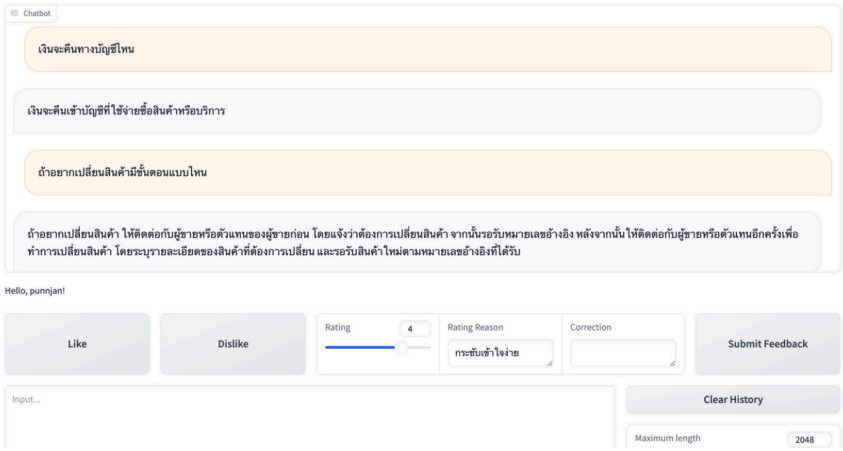

Figure 10: The model performs well in e-commerce Q&A, providing concise and clear answers.

However, as shown in Figure 11, the model struggled with gendered language in Thai, particularly with gendered particles. This issue was also observed in other state-of-the-art models, highlighting a common challenge in handling Thai gender distinctions.

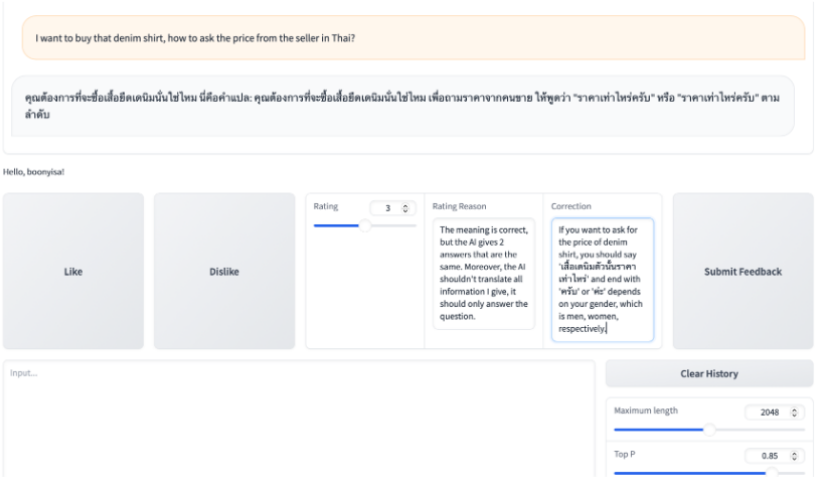

Figure 11: The model struggles with distinguishing gendered language in Thai, particularly with gendered particles.

Additionally, the LLM revealed that the unique ways of expressing time and numerical data in Thai pose challenges during translation, often resulting in inaccuracies and misrepresentation across languages. These subtleties make it difficult for models to handle temporal and numerical data with precision. Figure 12 illustrates how the model misrepresented time and numerical data.

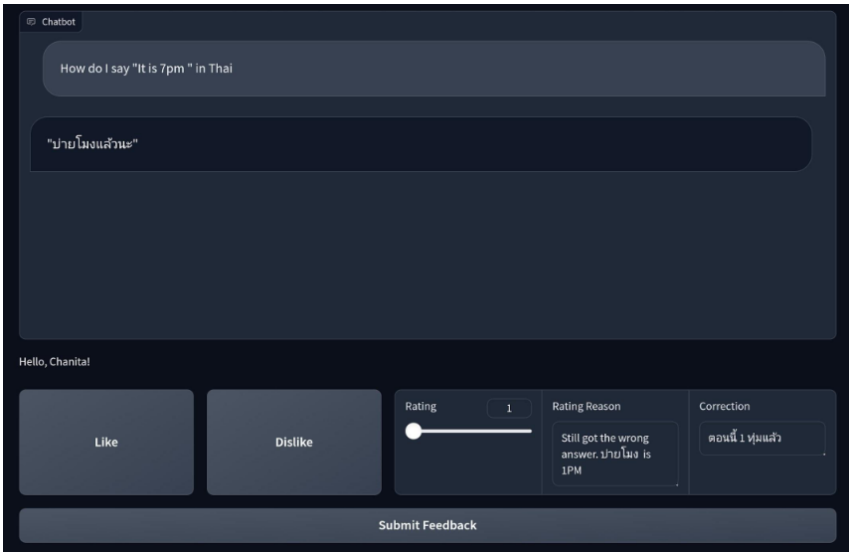

Figure 12: The model faces challenges in handling time and numerical data in Thai, leading to translation errors.

In summary, while our model performed well in e-commerce tasks, feedback highlighted its limitations in handling gender, politeness, and time-related expressions in Thai. These areas require further refinement.

9.2 Thai-Specific Evaluation Dataset: Xdata_Thai

In collaboration with Chulalongkorn University, we developed a Thai-specific evaluation dataset comprising 350 manually annotated samples. Created through ad hoc testing 6.1, peer reviews, and consensus-based annotations, the dataset ensures cultural and linguistic accuracy. It addresses low-resource language challenges by focusing on Thai idioms, slang, and formal tone distinctions. The task distribution, shown in Figure 13, highlights practical applications like e-commerce and idiomatic language, which are often overlooked in existing benchmarks.

To evaluate both pre-trained and instruction-tuned models, we formatted the tasks as continuation prompts in a 3-shots setting. The prompt format is detailed in Appendix 9.2.1. Table 6 presents the evaluation results, demonstrating the effectiveness of our instruction-tuned model in handling low-resource scenarios.

9.2.1 Creation of Evaluation Dataset

The development of the Thai-specific evaluation dataset was a meticulous, multi-step process aimed at ensuring cultural and linguistic precision. This dataset was designed to benchmark the performance of large language models (LLMs) in Thai, with a particular focus on cross-border e-commerce use cases. The dataset creation process consisted of the following phases:

Question Collection Process

1. **Ad Hoc Testing**: Thai university students were engaged to evaluate preliminary model outputs and suggest corrections. This process, described in Appendix 9.1, served as the foundation for question refinement.
2. **Peer Review**: Native Thai speakers in three independent groups reviewed the initial corrections and model outputs. These reviewers provided ratings, feedback, and alternative suggestions to enhance both linguistic and contextual accuracy.
3. **Final Selection**: The best answers were selected based on a thorough collaborative review process involving manual filtering and comprehensive quality assessment. Each question was then categorized by task type, as shown in Figure 13. This dataset includes an ex-

tensive collection of Thai idioms and proverbs, addressing nuanced aspects of language comprehension often neglected in existing benchmarks. Moreover, a significant subset comprises e-commerce-related questions, emphasizing practical application in cross-border trade scenarios.

Design Principles

To ensure the generation of high-quality questions, the following principles were adhered to [Johannes Welbl, 2017]:

- **No Yes/No Questions**: Questions were required to involve complex, meaningful tasks to challenge model reasoning.
- **Context Independence**: Questions were designed to be standalone, not requiring additional context for understanding.
- **Minimization of Ambiguity**: Careful attention was given to avoid vague or overly open-ended phrasing.

Distractor Design Strategy

High-quality distractors were critical for evaluating model robustness. The process of generating domain-relevant incorrect options followed these steps:

- **Leveraging Original Outputs**: Model-generated responses and corrections rejected during peer review were used as primary sources for distractors. Although suboptimal, these responses captured realistic errors, including both inaccuracies from the model and imperfect reviewer suggestions.
- **Supplementing with Generated Errors**: For missing distractors, errors were generated using an instruct-tuned version of XmodelLM [Wang et al., 2024a]. This model was trained on a dataset that excluded Thai-specific instructions, ensuring distractors were plausible yet domain-appropriate.
- **Cleaning and Replacement**: Repetitive, overly verbose, or low-quality responses were either cleaned or replaced. Replacements were drawn from actual e-commerce QA examples or other contextually relevant incorrect answers.

This rigorous process ensured that the dataset captures both linguistic subtleties and practical relevance, establishing it as a comprehensive benchmark for evaluating models in low-resource languages. To maximize its utility, we developed a structured evaluation framework encompassing task design, tailored evaluation metrics, and detailed data analysis, specifically suited for both pre-trained and instruction-tuned models.

9.2.2 Evaluation Task Design

To assess model performance, we designed the evaluation tasks as ABCD multiple-choice continuation prompts. The task design and evaluation methodology are as follows:

- **Randomized Options**: The answer options (A, B, C, D) are randomized for each question to mitigate positional bias.
- **Token-Based Matching**: The model generates a response continuation, and its predicted choice is determined from the first 10 tokens. This prediction is compared to the correct answer to calculate accuracy.
- **Few-Shot and and Chat-Adapted Evaluation**: The framework supports few-shot learning settings and optionally integrates chat templates to align more closely with the conversational nature of certain models.

The prompt format used for the evaluation is shown below:

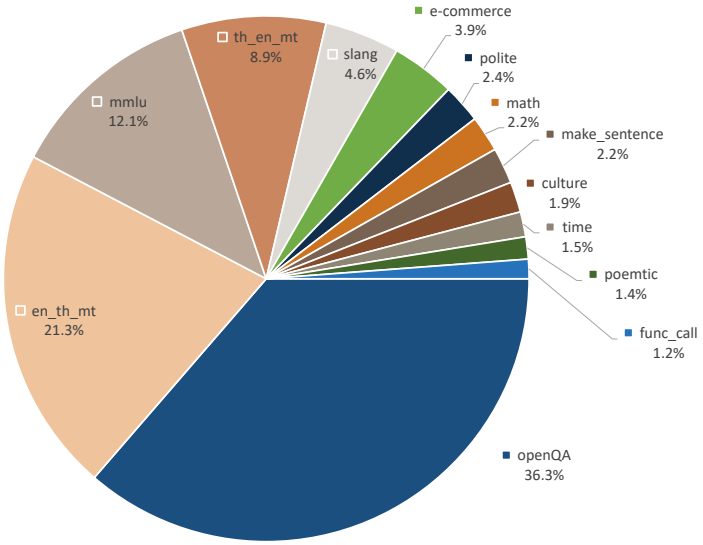

Figure 13: The distribution of task types in our evaluation set. Task types with fewer than 5 occurrences were removed.

```
The following are multiple choice questions (with answers) about Thai
      language knowledge.

Question: {{Question1}}
A. {{A}}
B. {{B}}
C. {{C}}
D. {{D}}
Answer: {{right_answer}}

Question: {{Question2}}
A. {{A}}
B. {{B}}
C. {{C}}
D. {{D}}
Answer: {{right_answer}}

Question: {{Question3}}
A. {{A}}
B. {{B}}
C. {{C}}
D. {{D}}
Answer: {{right_answer}}

Question: {{Question}}
A. {{A}}
B. {{B}}
C. {{C}}
D. {{D}}
Answer:
```

Figures 14 and 15 provide visual examples of these prompts. They illustrate how answer options are shuffled and how the generated continuations are compared with the correct answers for evaluation.

In the table below, we compare the performance of PolyLM-1.7B and Xmodel-1.5-1B on the Xdata_Thai evaluation dataset, using a 3-shot setting.

Model	**Xdata_Thai (3-shots)**
PolyLM-1.7B	0.228
Xmodel-1.5-1B	0.237

Table 6: Performance comparison on Xdata_Thai

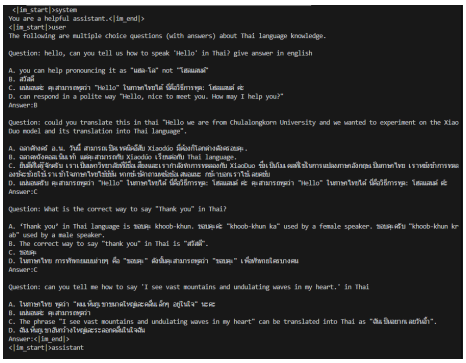

Figure 14: Example of a multiple-choice prompt incorporating the chat template.

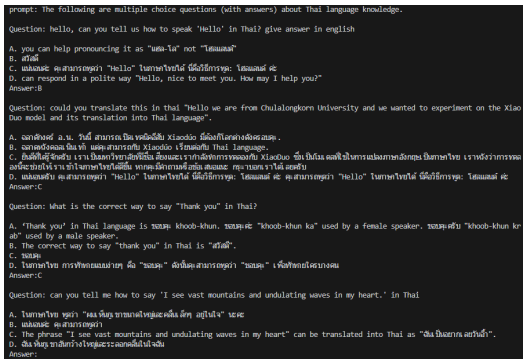

Figure 15: Example of a multiple-choice prompt without the chat template.

9.3 Detailed Evaluation Results

This section presents a comprehensive evaluation of the model's performance on multilingual tasks, focusing on Thai, Arabic, French, and Chinese. The evaluation results are reported using standard accuracy (acc) metrics, offering a clear comparison of the model's capabilities across these diverse languages.

Table 7: Performance on multilingual tasks (Thai, Arabic, French, Chinese).

Model	belebele_tha_Thai	xcopa_th
PolyLM-1.7B	0.2267	0.56
PolyLM-13B	0.2367	0.586
Xmodel-1.5-1B	0.2756	0.59

(a) Performance on Thai language tasks.

Model	arc_ar	hellaswag_ar	m_mmlu_ar	piqa_ar
PolyLM-1.7B	0.2173	0.2818	0.2288	0.5381
PolyLM-13B	0.2284	0.3296	0.2434	0.5653
Xmodel-1.5-1B	0.2344	0.3279	0.2454	0.5789

(b) Performance on Arabic language tasks.

Model	hellaswag_fr	m_mmlu_fr	paws_fr	piqa_fr
PolyLM-1.7B	0.3085	0.2458	0.548	0.5381
PolyLM-13B	0.4064	0.2602	0.539	0.5653
Xmodel-1.5-1B	0.37	0.2525	0.5325	0.5789

(c) Performance on French language tasks.

Model	arc_zh	xcopa_zh
PolyLM-1.7B	0.1957	0.5381
PolyLM-13B	0.2803	0.5653
Xmodel-1.5-1B	0.259	0.5789

(d) Performance on Chinese language tasks.